

A Comprehensive Literature Review of Penetration Testing & Its Applications

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Abstract: The PEN testing allows a PEN tester to check the functional aspects of a system that how much a system is vulnerable to the Network security & intrusion attacks & to see its defense mechanisms to counterpart these attacks. In this research paper we have conducted a literature review on the work done by the various researchers in the area of Penetration (PEN) testing. We have tried to review the various aspects related to the PEN testing. We have also studied the various tools used for PEN testing in terms of their utility, technical specifications, date of their release, Platform compatibility etc. various aspects related to the PEN testing. We have also studied the various tools used for PEN testing in terms of their utility, technical specifications, date of their release, Platform compatibility etc.

Keywords: *PEN Testing, Network Intrusion Attacks, Denial of Service, Network Security.*

I. INTRODUCTION

The Penetration (PEN) testing is all about checking your system to see that how well it is doing in terms of Network security & how well it is protected in terms of security countermeasures. It also includes ethical hacking of your system to see that how much vulnerable your system is to the Network intrusion attacks like Denial of Service (D.O.S.), etc. The PEN testing allows a PEN tester to check the functional aspects of a system that how much a system is vulnerable to the Network security & intrusion attacks & to see its defense mechanisms to counterpart these attacks. During PEN testing the functional aspects of a given system are tested to tell the testers that what functions they are exactly performing. A general goal of PEN testing is to break into your system & to check the loopholes into its defense mechanism to improvise the security lapses that are being left behind into your system security to ensure that the system must be protected from any exterior malicious codes that may try to breach the security of our system in order to gain unauthorized access to your data. It may have the possibility to misuse your data & may harm your computer network as well. The steps involved into a PEN testing include analysis of the entire organization's computer systems, secondly to look for the system vulnerabilities into it & third to identify their weak loopholes. The main objective of the PEN testing of system is to create & maintain information security of the proposed system in order to avoid such incidents that may violet the confidentiality & integrity of the secure data of an organization. In this paper, we have discussed about the PEN testing process for a system & its types. In the second part of the paper, we have discussed about the various PEN testing strategies. In the third & fourth part of the paper, we have discussed about various types of Pen testing & tools used in PEN testing. In the fifth part of the paper, we have discussed

about the research work done by the various authors into the area of PEN testing. In the sixth part of paper, we have concluded that PEN testing helps in framing & shaping the important aspects of information security parameters by identifying the vulnerabilities of a system in terms of network security in an accurate & efficient manner.

II. VARIOUS PEN TESTING STRATEGIES

In this section of the paper we have discussed about various types of PEN testing strategies depending upon the specific objectives achieved.

A. External PEN Testing Strategy

In external PEN testing of a system, it involves attack onto the organization's network, using procedures & codes that are defined & performed outside of the organization system's boundary using the Internet or Extranet. These tests usually begins with accessing the information that is available publicly about the client tally, then followed by the information about the network enumeration, then targeting the organization's externally visible services & network devices, like Domain Name Server(DNS), email server, firewall or Web servers etc.

B. Internal PEN testing Strategy

It is performed within the technical environment that is being present inside a system. This type of test basically involves imitating an attack onto the internal network structure of a system by an unsatisfied employee of an organization or by an authorized user of the system that is having the same access privilege, but they want to use the system for taking out their anger & frustration or to accomplish their motives of destruction. The focus of internal PEN testing is to understand that what would be the possibilities if the network parameters has been successfully breached by the attackers after the penetrating into you network. It describe the "what if?" for a network, if an authorized user goes bad, what damage he can do to your organization systems, when he has the ability to penetrate into the specific information resources about your organizations network.

C. Blind PEN Testing Strategy

It aims at imitating the actions & procedures followed by an attack onto a network into a real time environment. In this type of PEN testing, it occurs like a real life time hacking attempt, into which the testing team is supplied with either no information or limited information about an organization, before conducting the test. In this procedure, the PEN testing team uses the information about the organization that is

being available into their public domain to gather information about their target & to perform the PEN testing. The Blind PEN testing may discover a lot of information about the organization that was may not be known previously about the organization, like issues related to additional network access points, directly associated networks & confidential information about the organization that may be available into the public domain.

D. Double Blind PEN Testing Strategy

It is an extended version of the previous Blind testing. In the Double Blind PEN Testing strategy, the organization's IT staff & other resource persons that are responsible for the organization's security, don't know about the planned PEN testing activities, nor they are being informed about it. It is a very important aspect of PEN testing that, as it tests & maintains the organization's security parameters & responsive countermeasures that must be taken after intrusion detection. In this testing strategy only a few people of the organization knows about the testing activities & the involved procedures. In this the monitoring is done by the project manager who has the control of all the testing activities & organization's network intrusion detection parameters. These testing & response procedures can be terminated easily when the testing objectives of the Double Blind PEN testing strategy has been achieved.

E. Targeted PEN Testing Strategy

In this approach, both the organization's IT resource persons & people, the PEN testing team are being involved for carrying out the tests. In this the testing team will have clear idea about the PEN testing activities & information about the testing target onto which the network intrusion security breach is to be performed. It is a very efficient & cost effective technique as it focuses more onto the technical aspects of the design of the Network, rather than the countermeasures performed after intrusion detection. In this testing strategy, the targeted tests are being conducted & executed in a very less time & with minimum efforts as compared to the Blind testing.

III. TYPES OF PEN TESTING.

The various types of PEN testing depend upon the scope & requirements of an organization. Generally, there are three types of PEN testing techniques.

A. Black Box PEN testing

In this kind of testing, the PEN tester does not have any idea about the target system. Rather than the internal details, it is mostly emphasize is being given upon the gathering of information about the target network system. In this technique, the PEN testers know about the expected outcome but do not know about the possible outcomes. They do not know about that how the expected output will be generated. In Black Box PEN testing, the PEN tester does not bother about the internal structure of the network resources. In this type of testing, its main advantage is that it does not require any specific technical qualification to be inherited by the PEN tester & does not demand any specific technical knowledge to get it done.

B. White Box PEN testing

In this type of PEN testing, the PEN testers has been provided with all the type of information about the target system, which may include information about the schema, Structures of the target network system. It also involves source code details, Operating system details, IP addresses of target network. It is also called as Structural PEN tests or Open Box Pen testing. It performs the PEN testing by using code coverage & involves path flow testing, Data flow testing, etc. Its main advantage is that it covers all the main independent paths of a module & ensures 100% code coverage for the network fault detection. It is capable of finding the design errors in a network, which may occur due to the flaws in the logical flow of programs & between their actual executions of codes.

C. Gray Box PEN Testing

In this type of testing, the PEN tester is usually provided with limited or partial information about the internal detailed coding structure of the network that is to be targeted for a system. It is assumed that to be a kind of attack by an external user that have gained an unauthorized limited access to an organization target network infrastructure. Its main advantage is that it does not need the access to the source code. It is like coming without invitation & unbiased. It also draws a clear line of roles between the developer & a tester, thus allowing a very least or low risk of personal conflict between them. Further, one does need to provide the details about the internal structural information about the program or network to be tested.

IV. SOME COMMON PEN TESTING TOOLS.

As we know about that PEN testing involves all about the information collection, vulnerability & fault detection, exploitation vulnerabilities of a network or web application for detection of network attacks & then concluding their final results mentioning in to a final report preparation. This section provides information about various PEN testing tools that are being used by the PEN testers frequently. The descriptions of the PEN testing tools we have presented into tabular format in Table 1.

V. RELATED WORK ONTO PEN TESTING.

In this section we carried out a comparative review of the work done by the various authors in the area of PEN Testing & has shown the results in Table 2.

VI. CONCLUSION.

In this paper, we have carried out a review of the work performed in the area of PEN testing. We have also reviewed that how the PEN testing can be helpful in detecting the vulnerabilities that are present in our Network. We have also reviewed the various PEN testing strategies along with some common tools that are being used in the PEN testing of System. In the end, we can conclude that PEN testing is definitely a proven and efficient method to detect the flaws that are present in our System's security, and how we can ensure that to make our system foolproof to get protected from any kind of Network attacks.

TABLE I. SOME COMMON PEN TESTING TOOLS

Sr. No.	Testing tool	Platform	Coding Language	Purpose	Portability with Operating System	Ownership type	Release Date.
1	Hping.	CLI.	C C,C++,	Scanning of new ports + Remote access to PC.+ Finger Printing	Linux, Net BSD Unix OS, FreeBSD, Open BSD Unix OS.	Open Source.	November, 2005.
2	NMap.	GUI.	C,C++, Python, Lua	Network Scanning., Port Scanning, OS detection	Linux OS, Sun, Solaris, IRIX, Mac, Windows OS, HPUX, Net BSD & Open BSD	Open Source.	December, 2000.
3	Super Scan.	Cross Platform GUI based.	Perl, C++	Run queries like ping, local host name, lookups, who is, etc.+ Detection of open TCP/UDP ports+ Determination of services running onto TCP/UDP ports	Windows 2000, XP, Vista, 7.	Open Source.	April, 2003.
4	P0F Passive TCP/IP Stack Finger Printing Tool	Cross Platform Code.	C	Fingerprinting of OS.+ Detection of Firewalls	Linux, Net BSD, Free BSD, Open BSD Unix like OS, Windows, Solaris & AIX.	Open Source.	une, 2000.
5	X Probe Active Finger Printing Tool	GUI based Cross Platform Code.	Python	OS finger printing by Remote access. + Scanning of Ports. + TCP Stack Fingerprinting for identifying configuration attributes of new Network device.	Linux.	Open Source.	September, 1997.
6	Httpprint Web Server Fingerprinting Tool	CLI & GUI based Cross Platform	Python	Identification of Web servers to conduct web application based PEN testing.	Linux, FreeBSD, Mac OS, Win32 (Command line & GUI).	Open Source.	December, 2005.
7	Nessus Professional Vulnerability Scanner	Cross Platform GUI.	Nessus Attack Scripting Language	Remote security scanning.+ Discovery of Network vulnerabilities.	Free BSD Unix like OS, Linux, Mac OS, Apple, Windows and Solaris.	Freeware.	December, 2018.
8	GFI LanGuard Network Security Scanner	Cross Platform GUI.	VB. Script.	Provides detailed analysis of Network Status.+ Gives complete pictures of installed network devices , Hardware on Network, Software applications, Mobile Devices	Windows OS, Windows Server, 2003, 2008, Windows 2000 Professional Server.	Shareware	July, 2018.
9	Internet Security Scanner (ISS).	Cross Platform GUI.	C++	Enables system administrators to manage the security risks. + Provides speed measuring system based on RADAR. + Detection of Network Vulnerabilities	Windows 2000 Service Pack 4, Windows Server 2003, Windows XP Professional with Service Pack 1.	Shareware	August, 2013.
10	Shadow Security Scanner 7.55	Cross Platform GUI.	Visual C++.	Provides detection of holes in System's Security .+ Provides security algorithm that is based on patented intellectual core.	Unix, Linux, Free BSD, Open BSD, Net BSD, Windows 95/98/ME/NT/2000/XP.	Shareware	July, 2013.
11	Metasploit Project	Cross Platform CLI.	Ruby	Development & execution of exploit code using a remote target. + Tests vulnerability of Computer systems.	Unix, Windows, Linux	Proprietary.	January, 2019.
12	Brutus Sec. Tools	Cross Platform GUI.	C++.	Telnet,POP3,HTTP, FTP, RAS, IMAP password cracker.	Windows 9x/NT/2000.	OpenSource.	January, 2000.
13	Device 42 Network Scanner	Cross Platform GUI.	JavaScript	Is an agent less Network Scanner+ Discovers key dependences of Network using Native protocols like WMI for windows, SSH for links.	Windows, Linux	Shareware.	July, 2015.

TABLE II. A COMPARATIVE REVIEW OF THE LITERATURE WORK ONTO PEN TESTING

Sr. No.	Authors	Issues addressed	Key points	Merits	Demerits	Tools used
1.	Mathew D., et. al. [2016] [1]	A detailed study has been carried onto PEN testing for investigating its different aspects including tools, attack various methodologies, and defense strategies	Predominately used tools within the Kali Linux suite.+ Include hacking of Smartphone, hacking phones Bluetooth, Performed traffic sniffing, hacking WPA Protected Wifi, and many more	Has explained the concepts for better understanding of penetration test ing.+ presented different penetration tests using a private networks, devices, and virtualized systems and tools	Not Addressed	Kali Linux
2.	Hessa Mh., et. al. [2018] [2]	A Discussion is performed on the importance of PEN testing & its related factors consideration of its various components	Focusing on the various methods of PEN test, based on availability of information about testing type like black box, white box and gray box.+ Conduct of penetration tes-ting and documenting its outcome. +Test Analysis of Penetration Test.	Has mentioned about vulnerability assessment during PEN test in the security standards for IT e.g. ISO 27000 and + Coding of personal & professional ethics for persons involved in PEN test.	Not Addressed	NMap + BeEF+ Metasploit + Nessus + Cain & Abel
3.	Sandhya S., et. al. [2017] [3]	To assess a Website Secu-rity by perfor-ming PEN test using WIRESHARK	Discusses the importance of WIRESHARK + Use of Wire Shark as a Packet Sniffer	Revealing of flaws in the system's security at User authentication level for indicating that whether a website is secure or vulnerable.	Not Addressed	Wireshark tool
4.	R. Shanmuga priya et. al. [2013] [4]	A survey has been conducted on network security cyber crime technol-ogies using PEN testing	Discussed about the hacking techniques + Mention functions how it takes place in the network and the methods to be solved.	Discuss about various hacking methods and their prevention	Not Addressed	Not Addressed
5.	Nuno N. et. al. [2014] [5]	A study has been conducted onto PEN test for WEB Services using Web security scanning tools	Maximizing the effectiveness of vulnerability detection based on a Web service's out-puts. +Relaxation of black-box constraints for increasing the internal visibility of Faults within the tested Web service	PEN test play an important role in evaluating Web services security + Provides ability to correlate information provided by different types of requests+ Provides integrated support for detection of maximum number of vulnerabilities & minimal number of false positives.	Not Addressed	HP WebInspect + IBM Rational AppScan+ Acunetix Web Vulnerabilit y Scanner
6.	Xue Q. et.al. [2014] [6]	Proposing an automated method for solving problem like high costing & low efficiency during PEN test	Design & implement-ting an automated PEN Test tool called AEPT (Automatic Executing Penetration Testing).	Improve efficiency & accuracy of PEN Test + Greatly reduces cost of PEN test	Improvement in the design of prototype of the system is needed. + Enhancement of experimental setup & its verification is needed.	Java Language
7.	Chung Kuan C. et. al. [2018] [7]	A survey has been conducted on PEN testing tools & techniques needed to create defensive mechanisms for the new security threats posed in the Internet of Things (IoT) objects	Ensuring the security of IoT objects is not straight forward.+ Discusses three types of specialized PEN Testing at various levels like interface testing, transportation testing, and system testing for IoT objects.	Provides intelligent analysis methods that can be used to discover vulnerabilities for IoT objects	Not Addressed	Not Addressed
8.	Rakesh R. et. al. [2018] [8]	Implementation of an Automated hacking tool for web application security testing using Raspberry Pi Zero or Raspberry Pi Zero W with cost efficiency	To identify common web application security vulnerabilities + implementation of an Automated hacking tool for Web application PEN testing	discover additional and higher vulnerabilities	Needs lots of cash and a few internet application vulnerabilities might go undetected	Raspberry Pi Zero or Raspberry Pi Zero W
9.	Aleatha S. et. al. [2015] [9]	A detailed analysis has been performed theoretical vs. the practical classification for various PEN test frameworks & methodologies.	A survey is conducted for various PEN testing frameworks & methodologies, w.r.t ISSAF & OWASP's OTG	Many framework lacked domain coverage.+ Many were restricted in their applications.	Evaluation of selected frameworks w.r.t real-world case study is needed	Not Addressed
10.	Tao Lu, et. al.	A quantified comparative review	Detail the specific penetration test process in the Docker	Pen test detail for the specific penetration test process in the	Not Addressed	Not Addressed

	[2017] [10]	between Docker & traditional virtualization is made in terms of security aspect of Web applications	container environment	Docker container environment is studied		
11.	Shamal F. et. al. [2015] [11]	A survey has been made on several ethical dilemmas and dimensions faced by penetration testers.	about the ethics & dilemmas of PEN testing	1. have shown the differences associated with the dilemmas identified 2. have shown that how the study affects the ethical decision making before, during, or after a penetration test	Not Addressed	Not Addressed
12.	Raghaven dra P., et. al. [2015] [12]	discusses the stages involved in penetration testing	To draft a comprehensive and a systematic pen testing methodology for detection of malicious program's in real-time + to devise an effective scheme for management of a robust penetration testing environment	Penetration testing is an industry recognized term	Not Addressed	Not Addressed
13.	Trajce D., et. al. [2010] [13]	Introduces two methodologies for performing a physical PEN testing where the motive is to gain assets by using social engineering	Two methodologies were discussed for PEN testing using social engineering	Reduces the impact of the PEN test on the employees. + Provides meaningful & useful information on the security of an organization.	Not Addressed	Not Addressed
14.	Andrey P. et. al. [2008] [14]	Provides extent to Tainted Mode model that allows detection of intermodal vulnerabilities	Discusses an approach to vulnerability analysis by incorporating the advantages of PEN testing & dynamic analysis.	Proposes an enhanced Tainted Mode model that incorporates intermodal data flows.	main problem here is automatic form filling	Not Addressed
15.	V. Santhi, et. al. [2016] [15]	A investigation is made about different penetration testing tools in kali Linux,	Discusses about detailed PEN testing tools	Demonstrate the methodology used in hacking performed between the sender and the receiver	Not Addressed	Linux + Wireshark
16.	Sugandh S., et. al. [2013] [16]	Survey is conducted on methodologies & techniques involved in Vulnerability Assessment & PEN Testing (VAPT)	VAPT helps in identifying Cyber Threats and vulnerabilities under controlled circumstances	Proposed VAPT methodology is efficient, cost effective and assured assessment tool for current security arrangements analysis in an organization	Not Addressed	Not Addressed
17.	Yu Long Fu, et. al. [2014] [17]	Proposes an extended model of the classic IOLTS has been proposed for the security verification of protocol implementation	Discusses about the no negligible security properties with partial labels for identifying the security actions.	The proposed model helps the users to generate the possible transition sequences that contain the interactions between intruder and the protocol implementations.	Can't handle more complex experiments	Not Addressed
18.	Josip B., et. al. [2017] [18]	A discussion is made on approaches for automated security testing that are based on AI planning.	Use of AI planning offers representing attack models and security protocols in an effective manner & as bases for automating the security testing challenge	Web application attack models can be effectively shown as AI planning problem from which attacks can be obtained.	Granularity of actions and more reactive plan executions is still a challenge.	Not Addressed
19.	Irfan Y., et. al. [2017] [19]	A survey is done for identifying common types of network threats & the counter measures to prevent them.	Confidentiality, Integrity and Availability is achieved using VAPT + Gives over view of VAPT.	discusses about the different process & methodology for the Vulnerability Assessment & PEN Testing	Not Addressed	Not Addressed
20.	Harmande ep S., et. al. [2016] [20]	A survey is conducted for different types of PEN testing & to discuss about various tools & techniques for information gathering vulnerability assessment.	Provides a systematic review of all PEN testing Techniques & Frameworks	Provides a systematic review of all PEN testing Techniques & Frameworks	Not Addressed	Not Addressed
21.	Jai Narayan G., et. al. [2015] [21]	To prove VAPT as a Cyber Defense Technology	Complete life cycle of VAPT on systems has been defined.+ Open Source VAPT tools have been discussed like Metasploit, Nexus etc.	Emphasizes on VAPT has to be made as a compulsory for Cyber Defense	Not Addressed	Not Addressed
22.	Charles P.	To propose a systematic	View the system as collections	PEN tests can be described after	Not	Not

	P., et. al. [1989] [22]	approach for PEN testing of existing system using system security requirements	of objects that are going through one or more transactions, can be used to detect the system for vulnerabilities	examining the security controls, the sensitive objects for which they are used to protect & the functions used with these objects.	Addressed	Addressed
23.	Ramesh N., et. al. [2015] [23]	A study has been conducted on various methods used in the tracks covering and their future scope during PEN testing.	Discussed about Covering Track after PEN testing. +Described all the methods used in Covering Track.	Discussed about Covering Track after PEN testing.	Not Addressed	Not Addressed
24.	N. Priyanka, et. al. [2016] [24]	To propose a method for PEN testing to identify the vulnerabilities present in the Software Defined Networks.	To prove that the software defined networks are less secure for the online security threats.	Proposed that necessary step is taken in order to secure the network + Easy identification of attack surfaces, vulnerabilities, & other weaknesses of the Network from the attacker's point of view.	Not Addressed	Not Addressed
25.	Aileen G. B., et. al. [2011] [25]	To propose a methodology consisting of test preparation, test, & test analysis phase for PEN testing.	Test phase includes information gathering, vulnerability analysis, & vulnerability exploitation.	Selecting a penetration team is an important factor for the success of PEN test.	Doesn't offer absolute security level, but provides measure of security level.	Not Addressed
26.	Harshdeep S., et. al. [2017] [26]	A literature review has been done on the work related to security issues of wireless networks detected during their PEN testing.	Provides overview of PEN testing in Wireless security Networks	Proves that PEN testing is essential for validating the security mechanism of Wireless Security Networks.	Not Addressed	Not Addressed
27.	Konstantinos X., et. al. [2010] [27]	A survey has been conducted on the Role of modern Pen testers to find vulnerabilities in Organization's system network	Provides recommendations to overcome those vulnerabilities by hiring the services of Professional PEN testers	Identifies issues arising from pen-testers+ Mentions differences between their role & the industry demands.	Not Addressed	Not Addressed
28.	William G.J. H. [2009] [28]	To propose an approach for PEN testing using the input vectors of a web application to improve the Black box approaches.	A prototype tool SDAPT (Static and Dynamic Analysis based PEN testing) is developed to implement PEN testing.	Uses automated approach to detect the vulnerabilities in Web applications that contain SQLIA vulnerabilities. + SDAPT tool performs a through testing	Not Addressed	Not Addressed
29.	Daniel D. B., et. al. [2017] [29]	A survey has been conducted on the classification of various tools & models used in PEN testing	Proposed work focuses on mapping of the PEN test fields, identification of the application scenarios, usual tools & methodologies as per different contexts.	Helps testers to define to define within their testing scope.+ Defines how tools are used for vulnerability detection & definitions.	Not Addressed	Not Addressed
30.	Joel D. et. al. [2016] [30]	To propose a PEN test approach with strengths of existing PEN test approaches, to understand their utility & analyzing the security of a S/W program.	Proposed work investigates the applications of PEN testing to open source software during various stages of development of a WEB application.	Provides improvised security-based risk assessment+ rigorous test-driven threat modeling can be used to define PEN testing activities.	Not Addressed	Wireshark

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